KADI — Karnataka Analytics & Detection Intelligence

AI-Driven Crime Analytics & Visualization Platform for the Karnataka State Police · ಕಡಿ — “a link in a chain”

TEAM NAME	TEAM LEADER	TEAM SIZE	PROBLEM STATEMENT
KadiLabs	Adarsh Dwivedi	1	Challenge 02

59,985 FIRs	**85,429** evidence links	**578** repeat offenders	**100%** ground-truth recovery	**0.870** best model AUC	**13** Catalyst services

Deployed exclusively on Zoho Catalyst · every figure in this deck was read from the deployed API or a test run

Brief about the solution

The records are not the problem — the walls between them are.

The Problem today

FIRs sit in station-level silos — a station sees its own register, nothing else
No entity resolution — “Ravi Kumar”, “R. Kumar”, “Ravikumar D” are three people
SCRB gets fragmented extracts — no state-wide picture to act on
Reporting is retrospective — no early warning, no forecast
1. **Paper outruns typing** — a seizure memo is re-keyed by hand, or not at all

What KADI delivers

One living graph — every FIR joined to every other it shares evidence with
578 repeat offenders surfaced from 54,337 accused records
Eight ML models that each had to beat the rule a supervisor would use
Forecast · hotspots · investigation health each with a recommended action
1. **Bilingual grounded assistant** — EN / ಕನ್ನಡ, text or voice, always cited

59,985 FIRs analysed	**578** offenders resolved	**85,429** typed evidence links	**26,168** cases flagged for action	**100%** ground-truth recovery

One graph. 59,985 FIRs. Every link is evidence an officer can click.

A serial offender working across three districts stops looking like three unrelated cases — because the evidence that joins them is stored, typed, and shown on screen.

19 screens	**105** REST endpoints	**19** Data Store tables	**1,134** Kannada strings	**103** automated tests

Every number on this slide was read from the deployed Catalyst API or from a test run on this build — nothing here is illustrative.

The four challenges in the statement — answered

Each hurdle as the problem statement words it, what KADI does about it, and the evidence

1 Data silos & manual processes

"Records managed in independent silos, heavily reliant on Excel-based reporting rather than integrated, automated systems."

One graph over 19 relational Data Store tables, rebuilt nightly. 85,429 typed links join FIRs that no single station could ever see together, and every link names the attribute it matched on. Nothing is exported to a spreadsheet to be analysed.

85,429 typed links · **6** edge kinds · **0** spreadsheets

2 Lack of advanced analytics

"Absence of AI-driven approaches, leaving deeper behavioral patterns, social interactions and interconnected criminal networks undiscovered."

Eight ML models ship, and each had to beat the rule a supervisor would already use — twelve candidates did not and were cut. Behaviour, modus operandi and association are modelled and scored, not merely counted.

8 models shipped · **0.870** best AUC · **12** candidates cut

3 Information gaps at SCRB

"SCRB currently receives limited, fragmented information, hindering comprehensive state-wide analysis."

SCRB reads state-wide across all 31 districts and 298 stations — the same graph a station sees its own slice of, not a separate extract. Scope is enforced on every read, so one register serves every rank without a second pipeline.

31 districts · **298** stations · **1** register, six ranks

4 Reactive rather than proactive

"Policing remains largely reactive; without systematic exploration of emerging trends, investigators lack tools for proactive strategies."

Forecast, hotspots and pendency risk run ahead of the event. Five districts are pulsing red against their own history, 346 hotspots are clustered from location and time, and station pendency is predicted three months out.

5 districts pulsing red · **346** hotspots · **3 mo** forecast horizon

All four are answered in the deployed build, not on a roadmap. Every figure on this slide was read from the live Catalyst API while this deck was being written. The next slide maps the statement's six named capabilities onto the exact screen and model that delivers each one.

The six capabilities, requirement by requirement

Every clause of the statement, the screen that serves it, and a number read from the deployed build

REQUIREMENT	WHERE IT SHIPS	THE NUMBER
1a · District-level drill-down	Spatiotemporal Map — choropleth over satellite	31 districts · 298 stations
1b · Spatiotemporal clusters	Map — DBSCAN over location, layered hour × weekday	346 hotspots · 8 time-layered
1c · Emerging trend alerts	Command Dashboard — red-zone pulsing vs own history	5 pulsing · top district +145%
2a · Relationship mapping	Case-Linkage Graph — node canvas, 5 layouts	85,429 links · 6 typed kinds
2b · Repeat offender + MO	Offender Watchlist — MO across jurisdictions	578 repeat · 52,928 identities
2c · Association detection	Clusters — Louvain communities over the graph	100 clusters · largest 172 cases

REQUIREMENT	WHERE IT SHIPS	THE NUMBER
3a · Socio-economic correlation	Socio-economic Insights — per-capita, with p-values	+0.880 urbanisation, $p < 0.0001$
3b · Predictive risk scoring	React + Forecast — risk bands and horizons	0.870 AUC · 3 months ahead
3c · Anomaly detection	Investigation Health — deviation call-outs	1,803 cases · 6 stations flagged
4 · Pattern & trend discovery	Near-repeat, occasions and concentration curves	93% repeat rate · 400 m / 14 d
5 · Network & behavioural	Graph + agenda — mo_similarity is a first-class edge	shared MO ranked below offender
6 · AI/ML-driven intelligence	Eight models on QuickML and scikit-learn	12 candidates cut · fairness test

Nothing in the statement is unserved

Twelve requirements, twelve live screens. Capabilities 4 and 5 restate 1b and 2 at a different altitude, so they share the same analyses rather than duplicate them — the next slide shows that layer in full.

And one thing the statement does not ask for

Caste, religion and occupation are in the corpus and are excluded from every model by a test that fails the build. A platform that reads “social interactions” has to be told, in code, which social facts it may never learn from.

Pattern, trend and hotspot discovery

Capabilities 1b, 1c and 4 in full — how a hotspot is found, and what the concentration curve actually said

Spatiotemporal clusters — DBSCAN

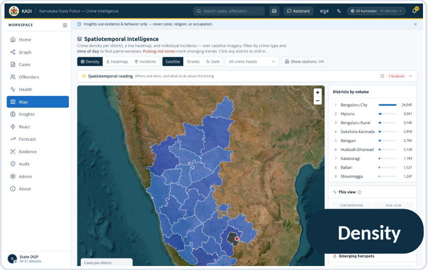
Density clustering over location at eps 0.004° (~450 m), minimum 8 incidents. 346 hotspots statewide; 8 survive when the same clustering is layered with hour and weekday, which is the set worth deploying against.

Near-repeat victimisation

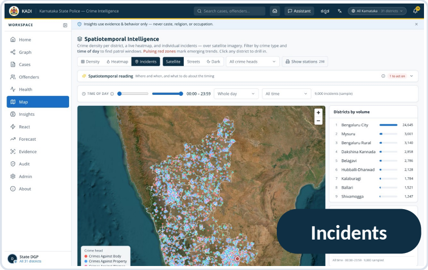
An incident with a prior inside 400 m and 14 days. The top cluster runs 70 incidents, 65 of them repeats — a 93% repeat rate at a median gap of 11 days. That is a place being re-targeted, not merely a busy one.

Occasion and day-class effects

Ten occasions scored against a 46.7 cases/day baseline across four day classes. New Year's Eve runs +34% and shifts the mix toward crimes against body. Festival effects are measured; civic and political ones are marked indicative.



Choropleth per district



9,000 dots by crime head

346 DBSCAN hotspots

8 time-layered clusters

5 districts pulsing red

26 open trend alerts

The concentration curve said something we did not expect

Gini across the 31 districts is 0.583. Across the 298 stations it is 0.047 — the busiest 10% of stations carry just 11.5% of the load, so resource cannot be moved between stations on volume alone.

0.583
districts — uneven

0.047
stations — nearly flat

0.377
clusters — the layer to deploy against

Tip : 0 is perfectly even, 1 is all load in one unit.

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Live analytics at a glance

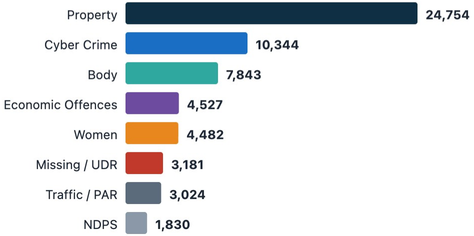
Six views of the same **59,985 FIRs** — every series on this page was read from the deployed Catalyst API, none of it illustrative

Monthly FIR volume + forecast

24 months actual · 3 projected · hold-out MAPE 7.8%

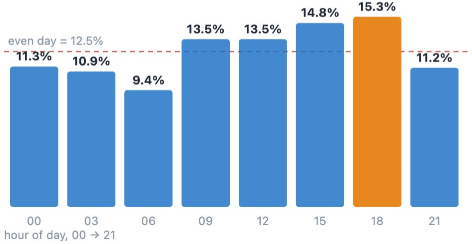
FIRs by crime group

Property and cyber carry 59% of the register



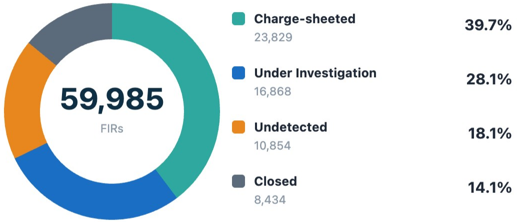
When crime happens

Eight three-hour shifts · 12.5% each if the day were flat



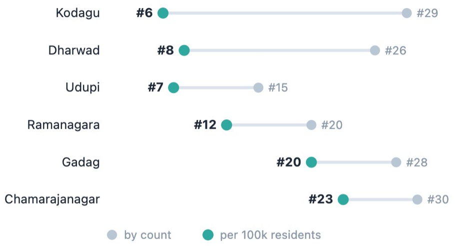
Case disposal

Every FIR accounted for · sums to 59,985



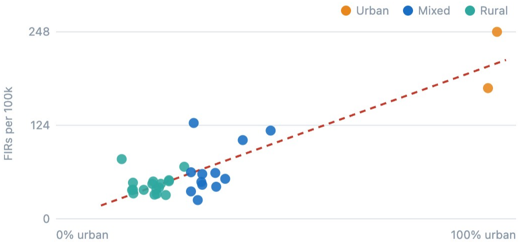
Rank by count vs rank per capita

What a raw-count map hides — same districts, two orderings



Urbanisation vs crime rate

n = 31 districts · Pearson r = +0.874, p < 0.0001



578 repeat offenders resolved	**85,429** typed evidence links	**100+** linked networks · largest 172 cases	**26,168** cases flagged for action	**100%** ground-truth recovery

Opportunities

How it differs, how it solves the problem, and the USP

1. How is it different?

Not a dashboard over counts

A graph over evidence. Every edge names the attribute that matched and the FIRs it matched on.

Entity resolution comes first

54,337 accused records resolved to 52,928 identities before anything is counted.

A model must beat the rule

Twelve candidates were built and rejected for not beating the heuristic they replace.

Fairness is a failing test

Caste, religion and occupation cannot reach a feature set — the build stops.

1. How will it solve it?

Breaks the silo, deliberately

A district officer sees cases linked into their district; a station still sees its own register.

Turns backlog into a queue

26,168 flagged cases ranked by severity, each with a recommended next action.

Warns before it happens

Station pendency +20% in 3 months, predicted at 0.870 AUC against a 0.701 rule.

Removes the retyping

Photograph a seizure memo; OCR reads it at 99% and files it against the case.

1. USP

Explainable end to end

Every edge, every score and every answer cites what it was built from.

Runs on Catalyst alone

Thirteen services. The 30-second request cap shaped the architecture, not a workaround.

Bilingual to the last label

1,134 Kannada strings, correctable in-product by the officers who read them.

Honest about its limits

The deck and the app both name what is not wired, and why.

The differentiator is not the chart library. It is that every claim on every screen traces back to the record it came from.

List of features offered by the solution

Twelve capabilities, all live in the deployed build

Case-Linkage Graph
Six typed edge kinds · 85,429 links · clickable “why linked” evidence

Entity Resolution
Rarity-weighted fuzzy matching · 54,337 → 52,928 · 578 repeat

Offender Risk
Glass-box factor breakdown · six horizons · name variants

Investigation Health
26,168 flagged, each with a reason and a recommended action

React — one ranked queue
Failing cases, active offenders and pulsing stations in one list

Spatiotemporal Map
Satellite · DBSCAN hotspots · hour × weekday · pulsing red zones

Forecasting
Statistical and ML side by side · MAPE 7.8% on a hold-out backtest

Socio-economic Analytics
Per-capita rates with p-values, so a weak signal reads as weak

Evidence Reading
OCR · barcode · vision · PDF and multi-page · files to the case

Bilingual Assistant
EN / ಕನ್ನಡ · text or voice · every answer cites its FIR numbers

Register & approvals
A station files an FIR; a supervisor approves it, before and after

Fairness & Audit
Protected attributes excluded by a failing test · every read logged

All twelve ship in the deployed build — none is a roadmap item.

Every figure on this slide was read from the live Catalyst API.

19	105	103	1,134	13
screens	REST endpoints	automated tests	Kannada strings	Catalyst services

Process flow

From a FIR register to an investigator acting on a lead

1 FIR intake

29 KSP tables · 59,985 FIRs over 43 months

2 Entity resolution

Blocking → rarity-weighted score → union-find merge

3 Link building

Six typed edge kinds · 85,429 links, each with its evidence

4 Analytics

Louvain communities · risk · health · DBSCAN · forecast

5 Delivery

Precomputed read-model, served to the SPA in under 100 ms

Where each stage runs — and why the platform forces the split

Catalyst Job · nightly 02:00 IST

A 15-minute budget. Entity resolution, graph build, Louvain, risk, health, DBSCAN and forecast — 24.6 s at 738 MB peak.

Function / AppSail · on request

Node 20, 512 MB. Reads what the Job already wrote and answers in under 100 ms. AppSail returns per-capita and forecast in ~135 ms.

Browser · instant

The SPA reads a precomputed read-model. No screen ever waits on a model, because no model runs behind an HTTP request.

The constraint that shaped everything: Catalyst Functions and AppSail both cap a request at 30 seconds — confirmed with the Zoho team in the Datathon workshop, and not raisable. Only Jobs get 15 minutes. So the pipeline runs nightly as a Job and the web tier only ever reads what it wrote. That single decision is why every screen loads instantly instead of waiting on a model.

24.6 s full pipeline run

738 MB peak memory, under the 512 MB cap

135 ms AppSail analytics

< 100 ms typical API response

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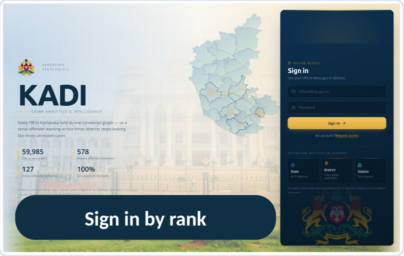
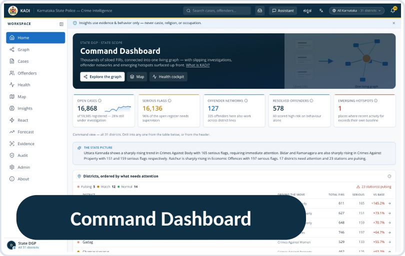
Wireframes of the proposed solution

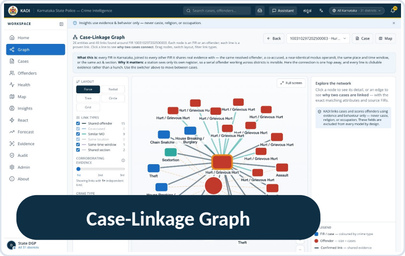
Not wireframes — the deployed build. Eight of nineteen screens, captured live at 3000 x 1880.



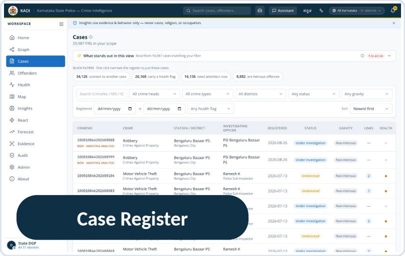
Six ranks, no password — scope follows the post



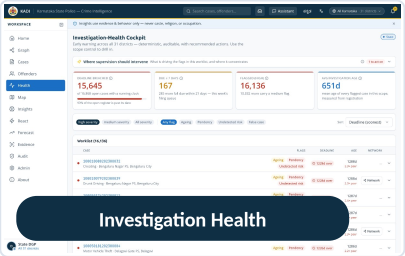
State picture in prose · districts ranked



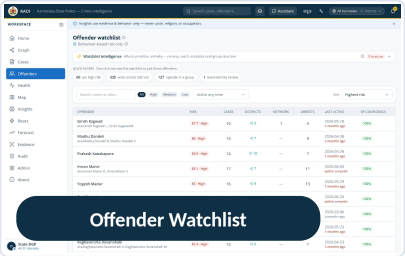
Ego-network · 5 layouts · 6 link filters



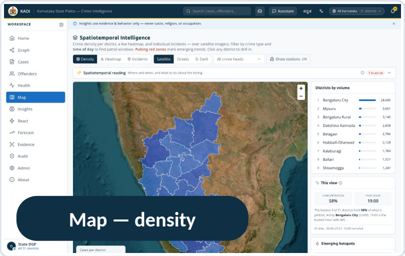
59,987 filed · 59,985 analysed



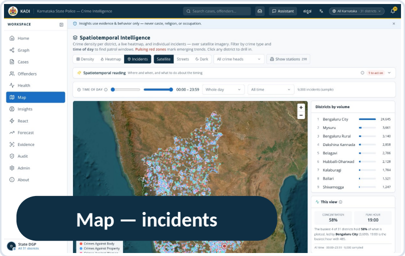
26,168 flagged, each with a next action



578 identities · glass-box risk



Choropleth per district over satellite

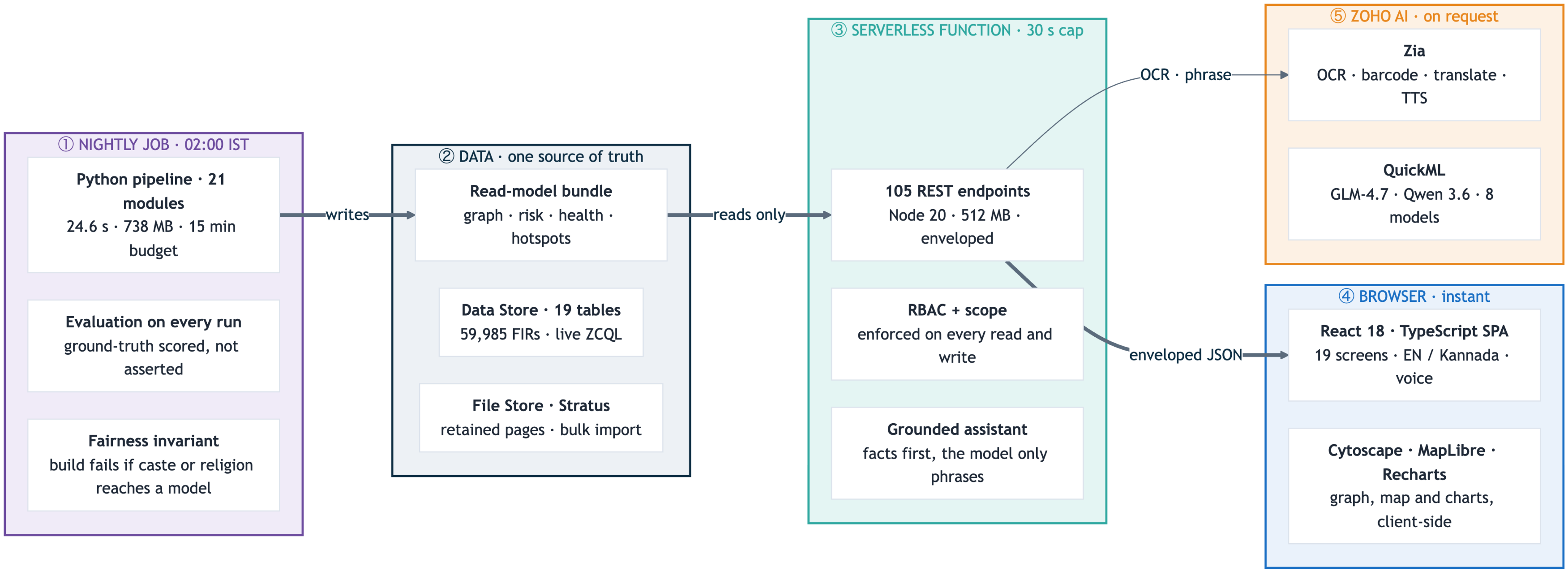


9,000 plotted dots, coloured by crime head

Every screen above is a live URL, not a mockup. Sign in by rank at the deployed link — no password — and the same scope rules an officer would get are the ones you get. The other eleven screens are on the snapshots slide and in the repository.

Architecture diagram of the proposed solution

Five tiers on Zoho Catalyst, drawn in the direction the data actually moves — the Job writes, the web tier reads



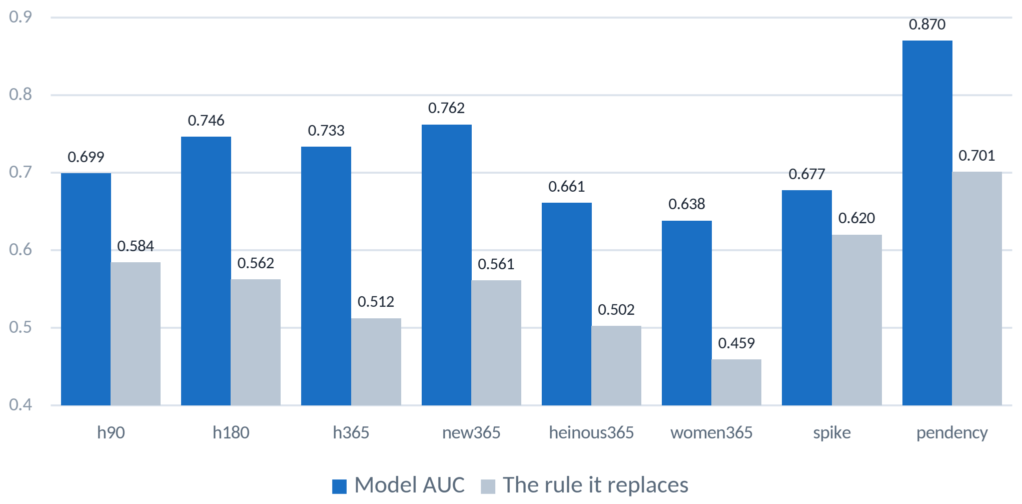
No heavy compute behind an HTTP request. Functions and AppSail cap at 30 s; only Jobs get 15 minutes.

The model never retrieves. It only phrases. Facts are computed against the register before any LLM is called.

The Eight ML models that ship

Every one had to beat the rule a supervisor would use instead — twelve candidates did not, and were cut

AUC on a time-ordered hold-out, against the rule each model has to beat



The rule that decided everything

For every target there is an obvious heuristic — the most recent offender reoffends, the busiest station falls furthest behind. Each candidate was scored against that rule on the same hold-out.

A model that cannot beat the rule is not a model. It is overhead.

Five tests a candidate has to survive

Beats the rule · overhead dressed as intelligence

Scale-free · has it only learned “big station is big”

Conditional · is one easy component carrying it

Poisson floor · is the spike just counting noise

Best-available baseline · was the rule chosen to flatter

What each model answers

h90 back on a new FIR within 90 days

h180 within 180 days — the default

h365 within a year

new365 next FIR in a district never worked

heinous365 next FIR recorded Heinous

women365 next FIR a crime against women

spike district × crime-head spike next month

pendency station pendency +20% in 3 months

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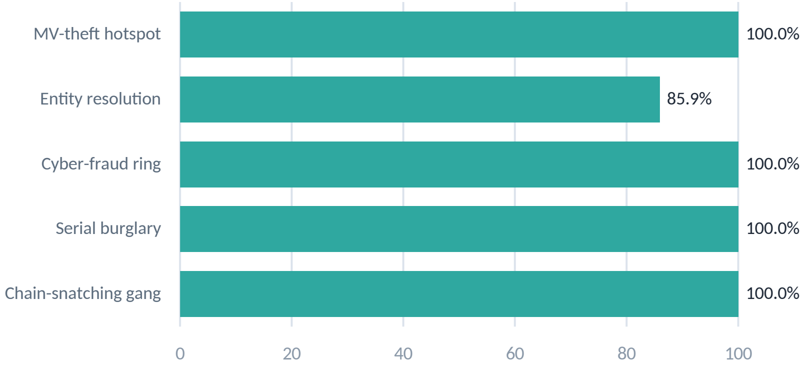
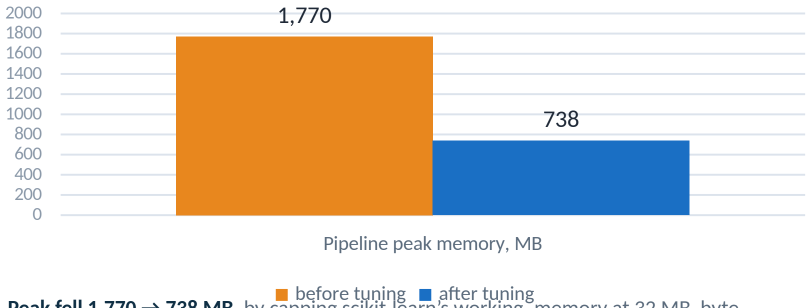
Prototype performance report / benchmarking

Measured on the deployed build — not estimated, and re-scored by the pipeline on every run

Ground-truth recovery seven patterns planted, then found again



Memory, before and after the one change that mattered



Peak fell 1,770 → 738 MB by capping scikit-learn's working_memory at 32 MB byte-identical output, zero speed cost, and it is what brought the run inside the 512 MB Job budget.

24.6 s full pipeline · 59,985 FIRs	**135 ms** AppSail analytics	**< 100 ms** typical API response	**7.8%** forecast MAPE, hold-out	**12.1 MB** graph payload, from 54.9	**99%** Zia OCR · 2.0 s	**127 ms** barcode · QR read

103 automated tests · 83 Node + 20 Python · 228 commits over 49 days. They cover the response envelope, RBAC at all three tiers, the scope refusal in both directions, evidence filing and retention, and a set of structural invariants — audit labels, route ordering, ZCQL limits, build output. Plus the fairness invariant, which fails the build if a protected attribute reaches a model's feature set. Seven patterns are planted by the generator at seed 2026 and re-scored on every run, so the recovery above is produced by the system rather than typed into this deck.

Who sees what — the access model

Six ranks, one register. Scope is enforced on every read and every write, and every privileged access is logged

Three tiers of scope, six+ ranks

State DGP	all 31 districts
SCRB Analyst	all 31 districts
Administrator	all 31, plus accounts
SP · district	one district
DSP · sub-division	one sub-division
SHO / SI · station	one station register

An account is pinned to its district and unit at sign-in; a query parameter cannot widen it.

Scope on the list AND the detail

A station SI reading the register sees 276 cases. Opening a case in another district is refused — 59,709 of 59,709 out-of-scope cases, verified live. The rule used to exist only in a comment.

Linked is visible, contents are not

A district officer sees that a case elsewhere is linked into their district — that is the product's thesis. Reading the seizure memo filed against it is not, so evidence stays stricter than the case.

The refusal is tested in both directions

Not only that an out-of-scope read is refused, but that an in-scope one still succeeds. A guard that refuses everything passes the first test and fails the job.

```
GET /cases/:id as SI —  before: 200 with named victims in 6
districts  ·  after: 0 allowed
```

Every privileged read and write is logged

Who, what, when, and a human-readable action — not a table of opaque codes. A record that can be read without a key is a record a supervisor will actually audit.

36 accounts **3** scope tiers **100%** privileged calls logged

Why this belongs in a crime-analytics deck: A platform that joins every district's register into one graph is, by construction, a platform that could show any officer anything. The scope model is the reason it does not. It was also the single worst defect found before submission case detail was unscoped while the list was scoped, so a station SI could open cases in every district sampled. Found by probing the deployment rather than by reading the code, and closed with a test in each direction.

Technologies used in the solution

Each chosen for a constraint, not for the list

Frontend

- React 18 · TypeScript · Vite
- Tailwind CSS
- Cytoscape.js + fcose — the graph canvas
- MapLibre GL — satellite and choropleth
- Recharts · Framer Motion
- TanStack Query
- pdf.js — lazy-loaded, multi-page evidence

What forced it
The graph had to stay interactive at five layouts and six link filters, with no round-trip per redraw.

Backend & API

- Node.js 20 · Catalyst Advanced I/O
- Express-compatible routing
- 105 REST endpoints, enveloped
- RBAC scoping on every read and write
- Persisted audit trail
- Grounded intent engine
- Enveloped errors — never a bare 500

What forced it
Advanced I/O takes an Express app unchanged, so the same 105 routes run locally and on Catalyst.

Data & ML

- Python 3.11 · pandas · NumPy · SciPy
- scikit-learn — TF-IDF, DBSCAN, NearestNeighbors
- networkx — Louvain communities
- RapidFuzz — rarity-weighted matching
- Shapely — district polygons
- QuickML — 8 tabular models
- statsmodels — trend and seasonality

What forced it
All of it runs inside the nightly Job. Nothing on this list is called from an HTTP request.

Platform & quality

- Zoho Catalyst · CLI 1.27
- ZCQL over 19 Data Store tables
- Zia · QuickML · SmartBrowz
- node:test · pytest — 103 tests
- Fairness invariant in the build
- GitHub — 228 commits, full docs
- Puppeteer — screens captured live

What forced it
The build fails if a protected attribute reaches a feature set. The claim is a test, not a sentence.

Nothing here is exotic — every choice traces to a Catalyst limit, the 30-second cap, the 512 MB Job budget, or the fairness rule. **The stack is deliberately boring so the evidence can be interesting.**

Catalyst services used in the solution

Thirteen in use — and an honest account of what is not wired

SERVICE	USED FOR	WHY THIS SERVICE
Web Client Hosting	Serves the SPA at /app	Same origin as the API
Serverless Functions	105-endpoint REST API	Advanced I/O takes an Express app
Job Scheduling + Cron	Nightly pipeline, 02:00 IST	Only Jobs get 15 minutes Functions cap at 30 s
AppSail	Python analytics service	Per-capita and forecast in ~135 ms, stdlib-only build
Data Store	19 tables · 59,985 FIRs	The FIR schema is genuinely relational
File Store	Retained evidence pages	Opt-in; deleted with its reading

SERVICE	USED FOR	WHY THIS SERVICE
Stratus	Bulk-import objects	Bulk-write reads its source from a bucket
Authentication	36 @ksp.gov.in accounts	An account is pinned to its district and unit
Zia	OCR · barcode · translate · TTS · STT	Trained models — no training data of ours
QuickML	GLM-4.7 · Qwen 3.6 · 8 tabular models	The only place a model runs at request time
SmartBrowz	Briefing export	Renders the print-ready briefing
Connections	OAuth for QuickML and Zia	Both reject anonymous calls

Not wired — four services attempted, diagnosed and stated rather than quietly skipped

Cache
401 PERMISSION_NEEDED

Adapter written and segment provisioned; writes from a deployed function are refused. Zero user impact — the KPI query recomputes in ~1 ms.

Zia object / identity
404 on every path

Every REST path tried returns 404 on this project. The vision model covers the same ground and is used in its place.

Zia face detection
ZIA_ERROR, always

The endpoint exists and errors on every image tried, including one with a face. Left off rather than shipped as a control that fails.

NoSQL · API Gateway
not provisioned

The read-model ships in the bundle, so NoSQL was never needed. Gateway was enabled once and disabled: with no routes it intercepted all traffic.

Estimated implementation cost

Free-tier today; the shape of a state-wide rollout

SERVICE	PROJECT USAGE	COST TODAY	AT PRODUCTION SCALE
Web Client Hosting	SPA + assets (~2 MB)	Free tier	Static hosting, negligible at KSP scale
Serverless Functions	2 functions · 512 MB · ~450k invocations/mo	Free tier	Scales on invocations; ~₹0.30 per 1k beyond free
AppSail	1 service · stdlib-only build	Free tier	Analytics on demand · ~135 ms per call
Data Store	19 tables · 59,985 FIRs · ~180 MB	Free tier	Real KSP volume ≈ 10 M FIRs — paid storage tier
Stratus + File Store	~8 MB import objects + retained pages	Free tier	Grows with retained evidence, opt-in only
Job + Cron	1 nightly run · 24.6 s	Free tier	One scheduled execution per day
Zia + QuickML	OCR, translate, TTS, STT, LLM, 8 models	Consumption	Priced per call — the only usage-linked line

Rs 0 / month today

The whole prototype fits inside the Catalyst free tier : 59,985 FIRs, the nightly pipeline, and every screen.

Only Zia and QuickML are usage-priced.

At real KSP scale

~10 M FIRs, against 59,985 here

~167× the corpus this runs on

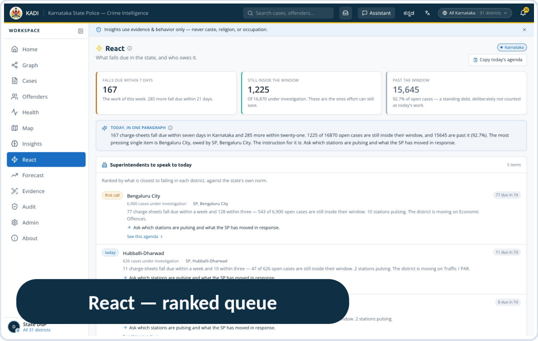
Nightly still one Job the pipeline is linear

Storage the only line that changes tier

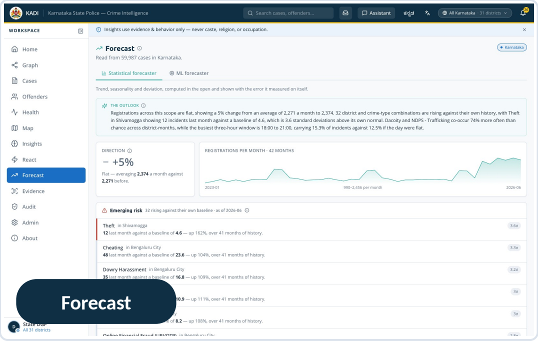
What actually changes at 10 M FIRs. One line: Data Store moves to a paid storage tier. The nightly Job stays one Job, the read-model stays one bundle, and the three request tiers keep the same shape the pipeline is linear in corpus size and already clears 59,985 FIRs in 24.6 s. There is no re-architecture priced into this table because none is needed.

Snapshots of the prototype

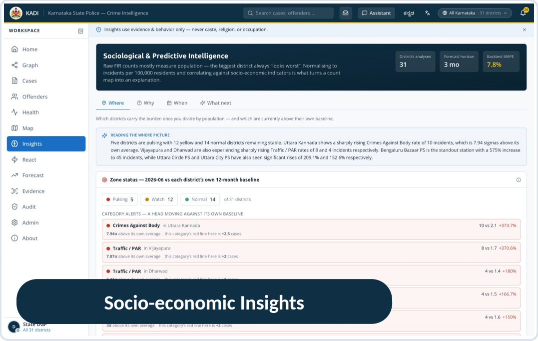
The other six screens : Every one captured from the deployed Catalyst URL



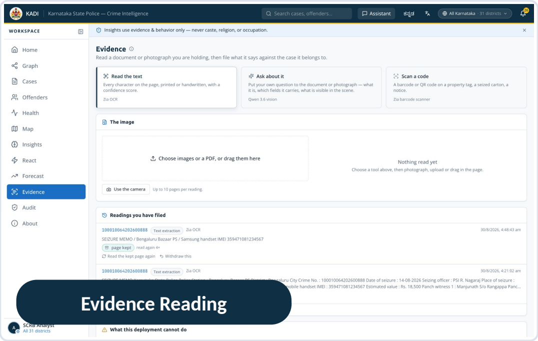
Failing cases, active offenders, pulsing stations



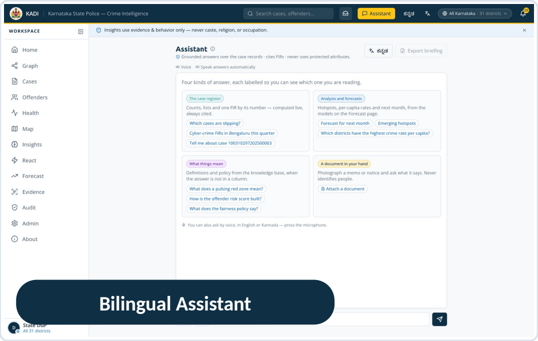
Statistical and ML forecaster, each with its error



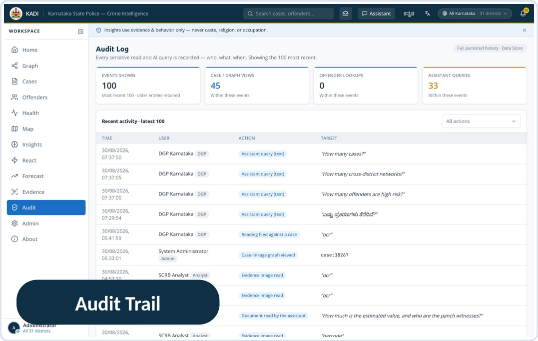
Per-capita rates and rank shift, with p-values



OCR, barcode and vision — filed against the case



Four kinds of answer, each badged with its source



Every privileged read and write, in plain language

Links

Everything is publicly accessible and independently verifiable

LIVE APP
Deployed solution — Catalyst
Web Client Hosting · sign in by rank, no password needed
https://kadilabs-60078029367.development.catalystserverless.in/app/

VIDEO
Demo video
Problem overview · working prototype · key workflows
https://drive.google.com/drive/folders/1WY3KHg1WOEnSNTBXGmTtH2ZoJM1y4cLJ?usp=sharing

SOURCE
GitHub repository
228 commits over 49 days · full source, README, docs
https://github.com/adarshcod30/Kadi

LIVE API
AppSail analytics service
Live JSON — per-capita rates and correlations
https://kadi-appsail-50043957273.development.catalystappsail.in/analytics/socio

Verify the whole claim in five minutes

1. **Sign in as State DGP**
   no password — pick the rank
2. **Open a case on the Graph**
   click an edge; it names what matched
3. **Sign back in as a station SI**
   the same case is now refused

Built in the open, entirely on Catalyst

228	49	103	16
commits	days	tests	docs

Client, API, Python analytics, nightly job, database and file storage are all Catalyst services.

Additional details & future development

Six items, in priority order — each one a swap behind an interface that already exists

1 Read the register from Data Store — NEXT

The 59,985 FIRs are already in Data Store and queryable via ZCQL — ?source=datastore proves it live. The default path serves a precomputed bundle for sub-100 ms response; swapping it sits behind the existing store interface.

2 Native-speaker Kannada review — READY

The instrument exists and is live: any officer can correct a string in ten seconds and it applies for everyone, attributed and reversible. What remains is the review itself, not the tooling.

3 Incremental recompute on insert — DESIGNED

The pipeline rebuilds nightly in 24.6 s. Signals on FIR insert would move that to incremental, so a case registered at 14:00 is linked by 14:01 instead of by 02:00.

4 Extend the graph — MODEL READY

Vehicle registrations, phone and IMEI, and bank accounts as additional typed link kinds. The edge model already supports them; this corpus does not yet carry the columns.

5 Retain every page, read the text layer — SCOPED

A multi-page reading keeps only its first page today, and PDFs are rasterised rather than having their existing text extracted. Both are storage and parsing, not architecture.

6 Configure API Gateway routes — BLOCKED

Enabled once and disabled again: with no routes configured it intercepted all traffic. It needs a route table before it can go back on — stated here rather than left off the list.

Impact at real KSP scale

The rollout question is storage tier, not architecture.

~10 M — FIRs at production scale, against 59,985 here

~167× — the corpus this already runs on in 24.6 s

0 — rewrites needed — the pipeline is linear in corpus size

Nothing on this list is a rewrite. Every item is a swap behind an interface that already exists — the store interface, the translation table, the edge model, the retention flag, the route table. That is what shipping the boring version first buys you: the next six things are configuration and data, not a second architecture.

The dataset, the fairness rule, and the finding

Synthetic FIRs, generated against a real skeleton and what the analysis surfaced

The corpus

59,985 FIRs analysed · 43 months

31 / 298 districts / police stations

54,337 accused → 52,928 identities

85,429 typed evidence links

7 ground-truth patterns planted

Synthetic FIRs only — no real case or person appears. Generated against the real KSP schema, geography and statistics.

Fairness is an invariant, not a policy

Caste, religion and occupation never enter entity resolution, linkage, risk scoring or any prediction and the guarantee is executable.

```
PROTECTED_COLUMNS = {ReligionID,
  CasteID, OccupationID, …}
raise ValueError(FAIRNESS_VIOLATION)
```

A unit test fails the build if any protected column reaches a model's feature set. The claim is a test, not a paragraph of policy.

What the analysis surfaced

Raw counts mostly measure population. Normalising to incidents per 100,000 residents changes the map entirely.

Kodagu	30th by count	6th by rate
Dharwad	23rd by count	7th by rate
Tumakuru	10th by count	24th by rate

335 FIRs looks unremarkable beside Bengaluru City's 16,895 — until you divide by Kodagu's 648,787 residents. A count map would never surface it.

Socio-economic correlation across all 31 districts, with p-values so a weak signal reads as weak

+0.871
population density
$p < 0.0001$ · strong

+0.880
urbanisation
$p < 0.0001$ · strong

+0.546
literacy
$p = 0.0018$ · moderate

5.4×
urban vs rural rate
163.6 against 30.1 per 100k

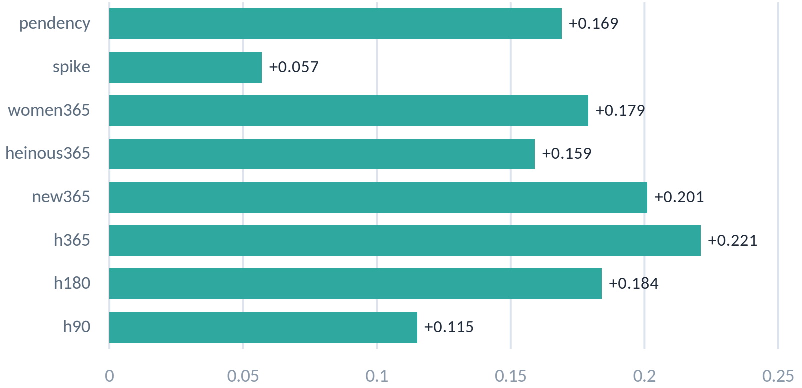
Stated openly: the generator weights urban crime upward, so the urbanisation correlation is partly circular.

The method is sound and runs unchanged on real KSP data — but on this corpus it is confirmation, not discovery. We say so here rather than let an evaluator find it. The same honesty runs through the app: three Zia image services do not answer on this project, and the Evidence screen says which and why instead of mocking them.

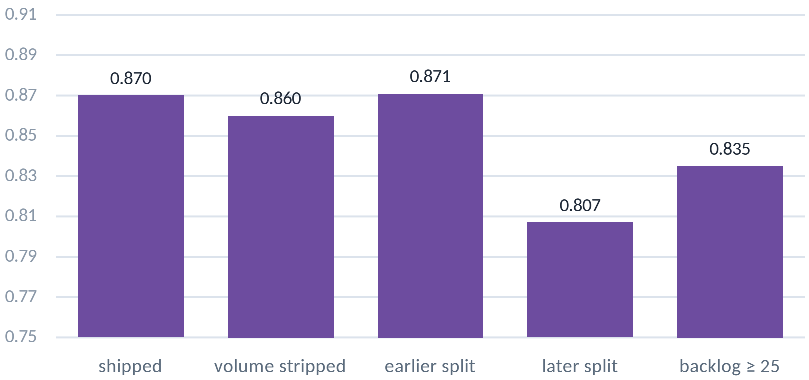
How the models were evaluated

The margin over the rule, and what happens to the best model when the ground is moved under it

Margin over the rule — every model clears it



Station pendency, re-scored five ways



Why the margin is the number that matters

An AUC of 0.87 sounds strong until you learn the obvious heuristic scores 0.70 on the same split. The bar that counts is the gap, and twelve candidates were cut for having none.

The robustness checks are the honest half

Strip every absolute volume feature and it holds at 0.860 — so it has not merely learned that a big station is big. Move the split earlier or later and it holds. That is what makes it deployable.

Read the spike margin honestly. +0.057 is thin. The target — 40% above the trailing mean — is easier to hit on a small base, and the Poisson floor test is what keeps that from being mistaken for signal.

Karnataka State Police
Government of Karnataka

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Datathon 2026

Technology partner ZOHO

THANK YOU